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A

Power – 12V, solar and USB inputs, 18650 charger, rails

File: power.kicad_sch

ESP32, programming, buttons, RTC, LED strip, buzzer

File: mcu.kicad_sch

RS-485 and uplink

File: comms.kicad_sch

B

AgriNode HUB-C (4G hub)

One board per hub type: HUB-W (Wi-Fi), HUB-L (LoRa), HUB-C (4G LTE module board), GW-LAN (LoRa to Ethernet)

Core: ESP32-WROOM-32E, 12 V / solar / USB-B inputs, 18650 + TP4056 + power path, MIC29302 3.3 V, DS3231M RTC

USB programming (CH340C, auto-reset) + PROG header, RESET and BOOT/PAIR, WS2813 LED-strip header, buzzer

Optional RS-485 (Modbus) – leave unpopulated if not needed

Source of truth: design/hubs.py – regenerate with build.py <variant>

C

D

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F

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AgriNode

Sheet: /

File: agrinode_hub_c.kicad_sch

Title: AgriNode HUB-C (4G hub) – Overview

Size: A3

Date: 2026-09-27

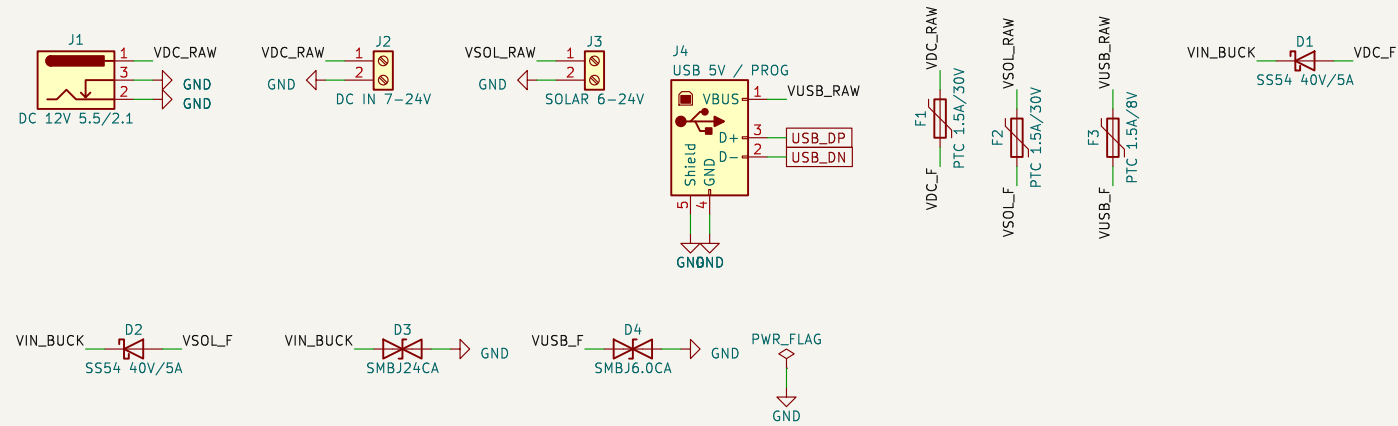
Rev: A

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

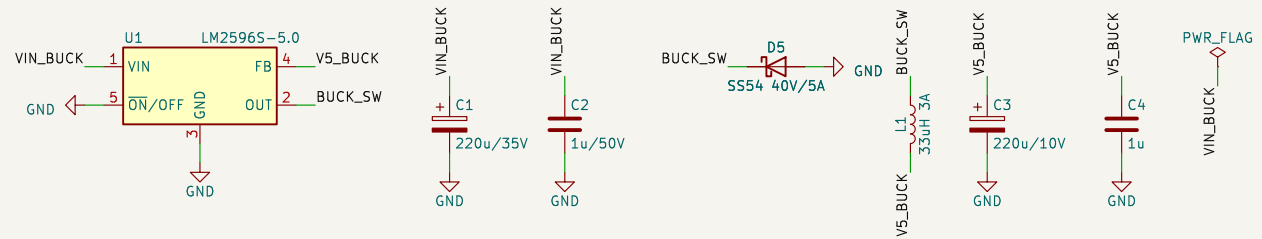
Id: 1/4

Inputs: 12 V adapter (jack + terminal), solar 6–24 V, USB 5 V

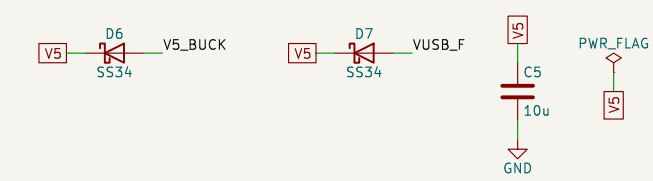
each input fused and reverse–polarity protected; any combination may be connected



5 V buck (LM2596S–5, 7–40 V in, 3 A)

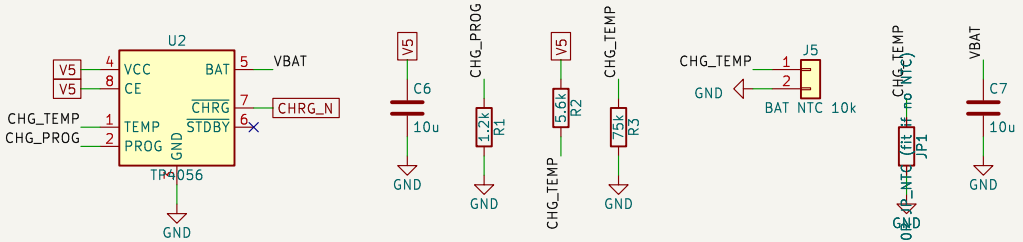


5 V bus (diode–OR of buck and USB)

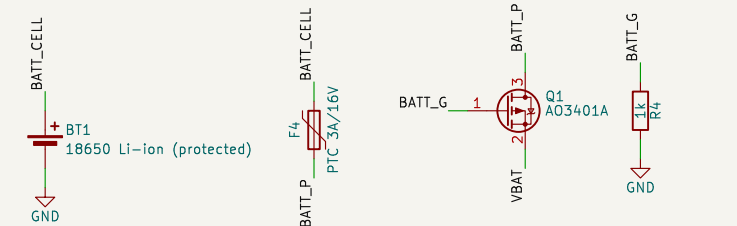


18650 charger (TP4056, 1 A) with battery temperature guard

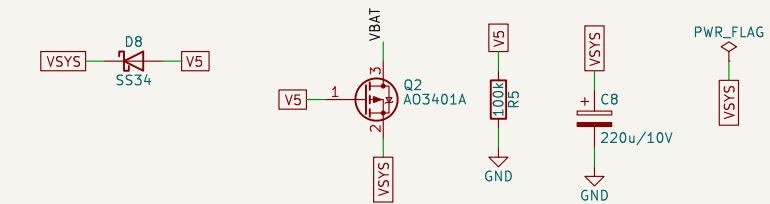
NTC 10k B3950 on the cell; charge only 0..45 C. No NTC: fit JP_NTC (TEMP to GND)



18650 cell: holder, fuse, reverse–insertion protection

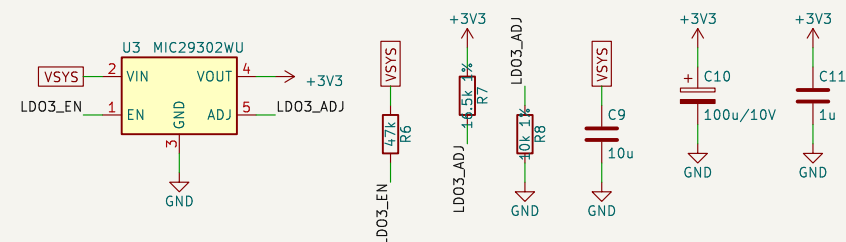


Power path: 5 V bus when present, battery otherwise (no gap)



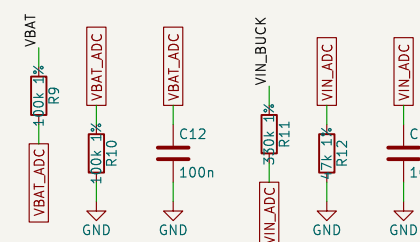
3.3 V rail (MIC29302, low dropout, 3 A)

$V_{out} = 1.24 \times (1 + 16.5k/10k) = 3.29 \text{ V}$

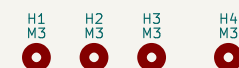


Supply monitoring (ADC1)

VBAT: 100k/100k VIN: 330k/47k (24 V -> 3.0 V)



Mounting holes (M3)



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AgriNode

Sheet: /Power – 12V, solar and USB inputs, 18650 charger, rails/

File: power.kicad_sch

Title: AgriNode HUB–C (4G hub) – Power – 12V, solar and USB inputs, 18650

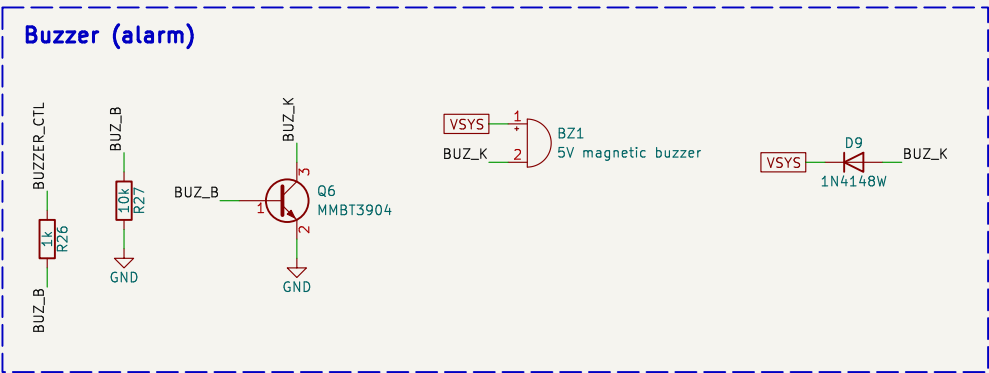
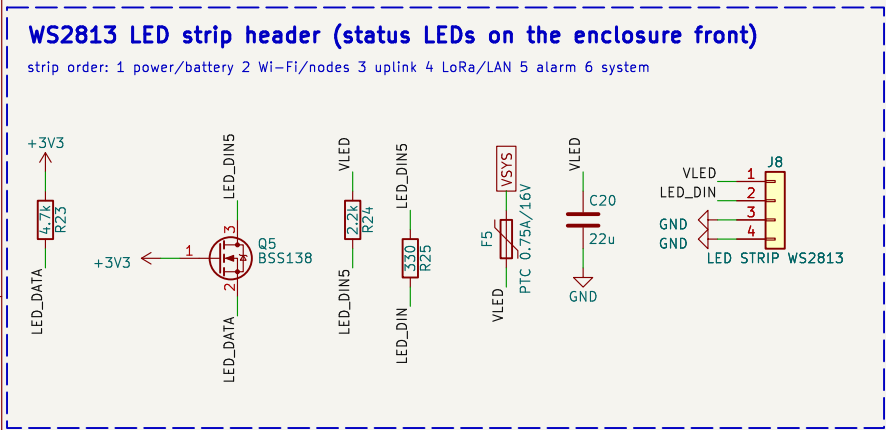
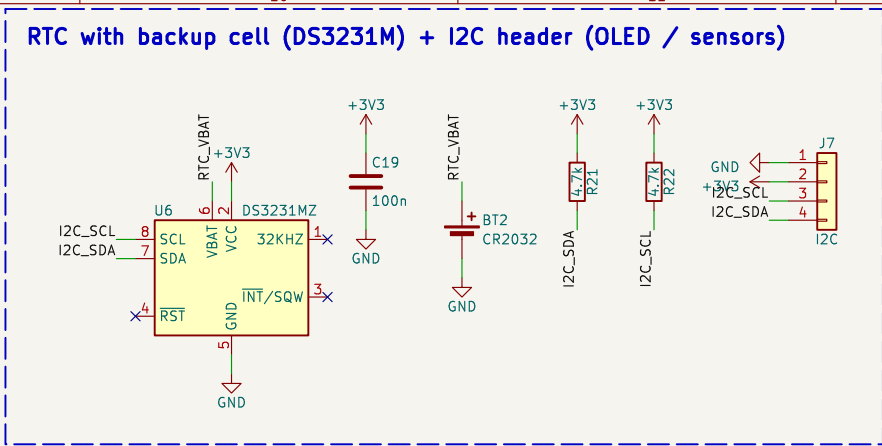
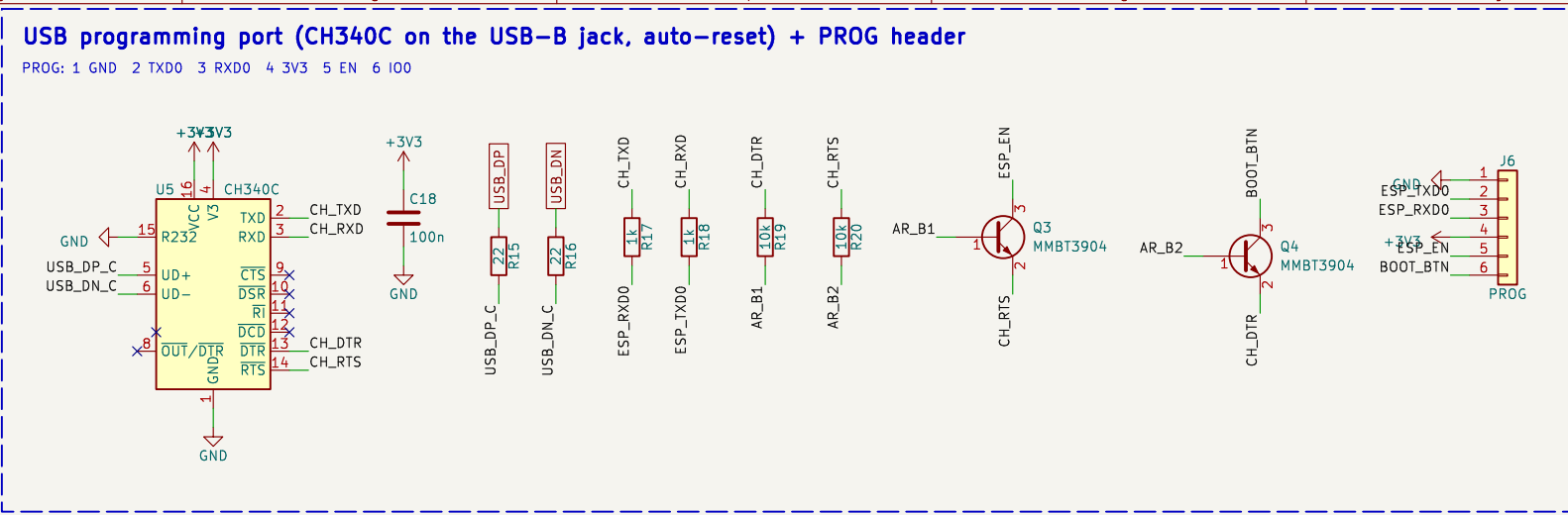
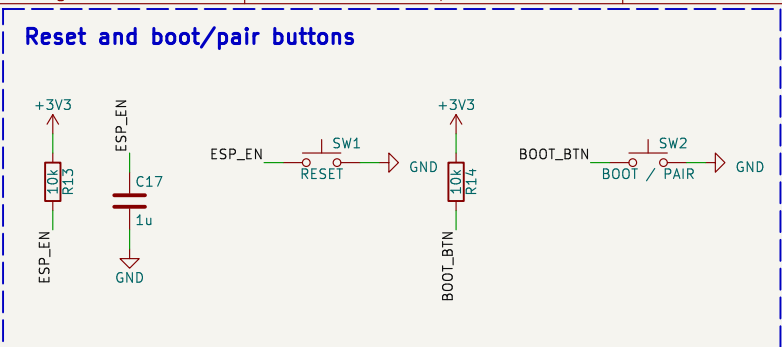
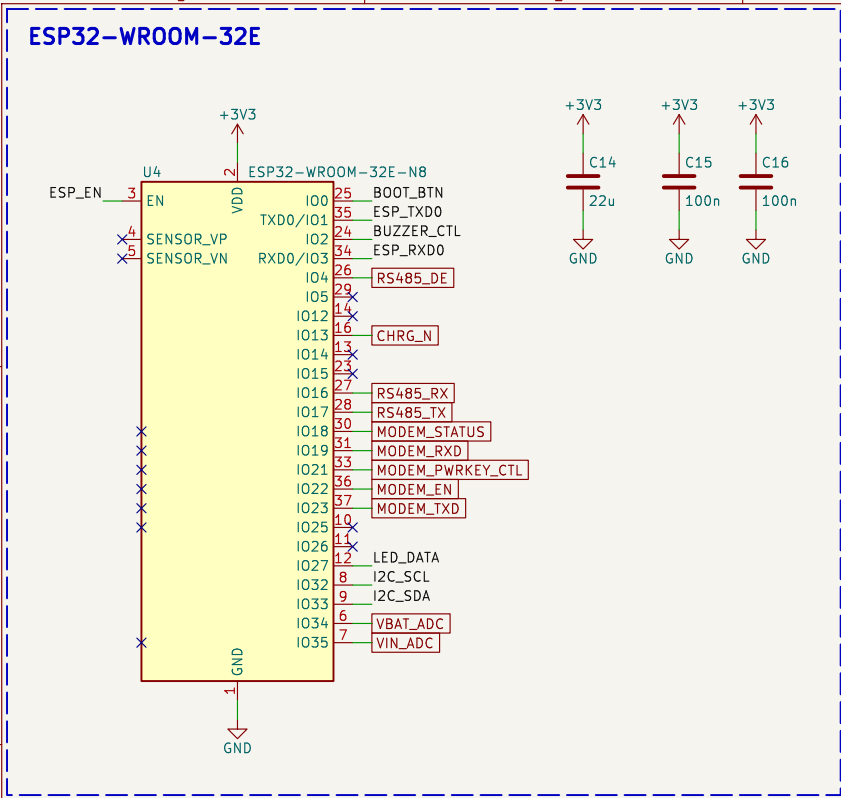
Size: A3

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Rev: A

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Id: 2/4



The schematic diagram illustrates the RS-485 / Modbus RTU interface for wired industrial sensors, supporting distances up to 1 km. The circuit is divided into four main sections:

- Microcontroller (U7):** A MAX3485ESA transceiver is connected to a +3V3 supply (pin 8) and ground (pin 5). It interfaces with an RS-485 module (JP2) via pins 1 (RS485_RX), 2 (RS485_DE), 3 (RS485_DE), and 4 (RS485_TX). The transceiver's output pins 6 (RS485_A) and 7 (RS485_B) are connected to the RS-485 module.
- RS-485 Module (JP2):** The module is connected to the transceiver and the RS-485 network. It includes a 100nF capacitor (C21) connected to +3V3 and ground, and a 10k resistor (R28) connected to ground.
- RS-485 Network:** The network consists of two twisted-pair lines, RS485_A and RS485_B, which are terminated at both ends. The termination is achieved using 120Ω resistors (R29 and R31) connected to ground.
- RS-485 Connector (J9):** The RS-485 network is connected to the RS-485 connector (J9) via pins 1 (RS485_A), 2 (RS485_B), and 3 (GND).

